



Predicting Customer Churn in a Telecommunications Dataset

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DSC630 Predictive Analytics

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March 8, 2026

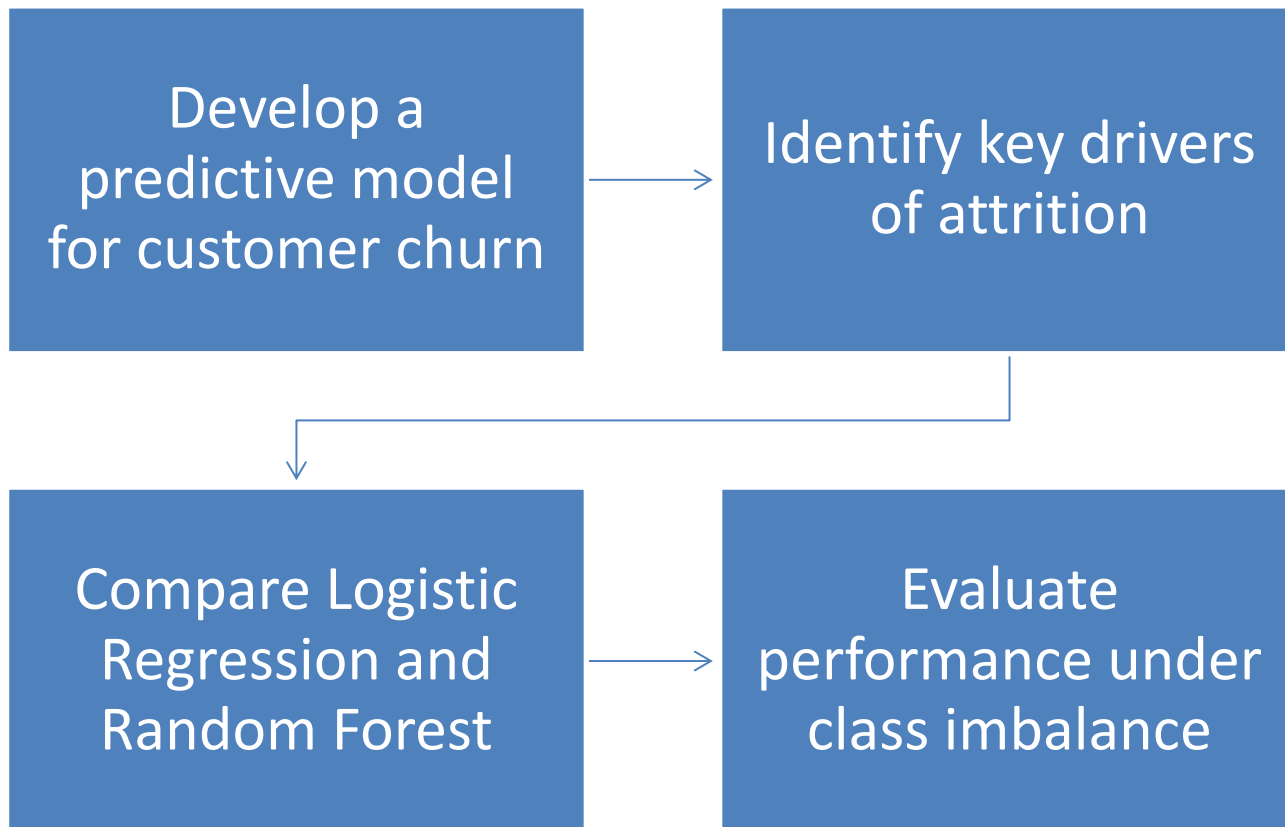


Why Customer Churn Matters

- **Customer attrition** leads to revenue loss
- **Acquiring new customers** is more costly than retaining existing ones
- Retention requires **early identification** of at-risk customers
- **Predictive analytics** supports proactive intervention



Research Objective



Dataset Overview

- Telco Customer Churn dataset (Kaggle)
- **7,032** records after preprocessing
- **26%** churn rate
- Demographics, services, billing features
- Binary classification problem



Data Preparation

- Converted TotalCharges to numeric format
- Removed missing values
- Dropped customerID field
- Applied one-hot encoding
- Standardized numerical features
- 80/20 stratified train-test split



Modeling Approach

- Models Evaluated
 - Logistic Regression
(baseline, interpretable)
 - Random Forest
(nonlinear ensemble method)
- Evaluation metrics
 - Accuracy
 - Precision
 - Recall
 - F1 Score
 - ROC-AUC



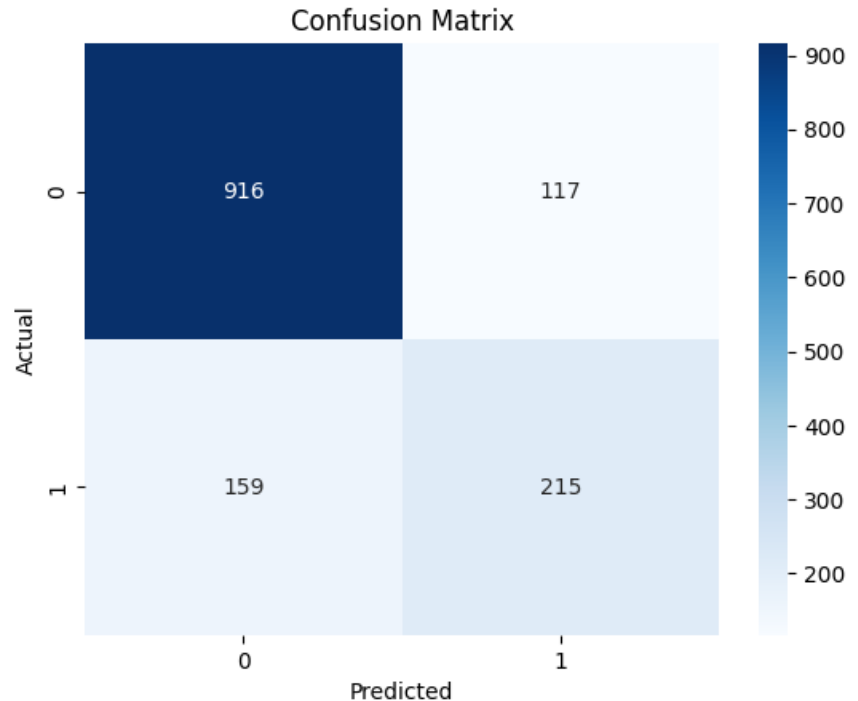
Logistic Regression Performance

Metric	Value
Accuracy	80.4%
Precision	0.65
Recall	0.57
F1 Score	0.61
ROC-AUC	0.84



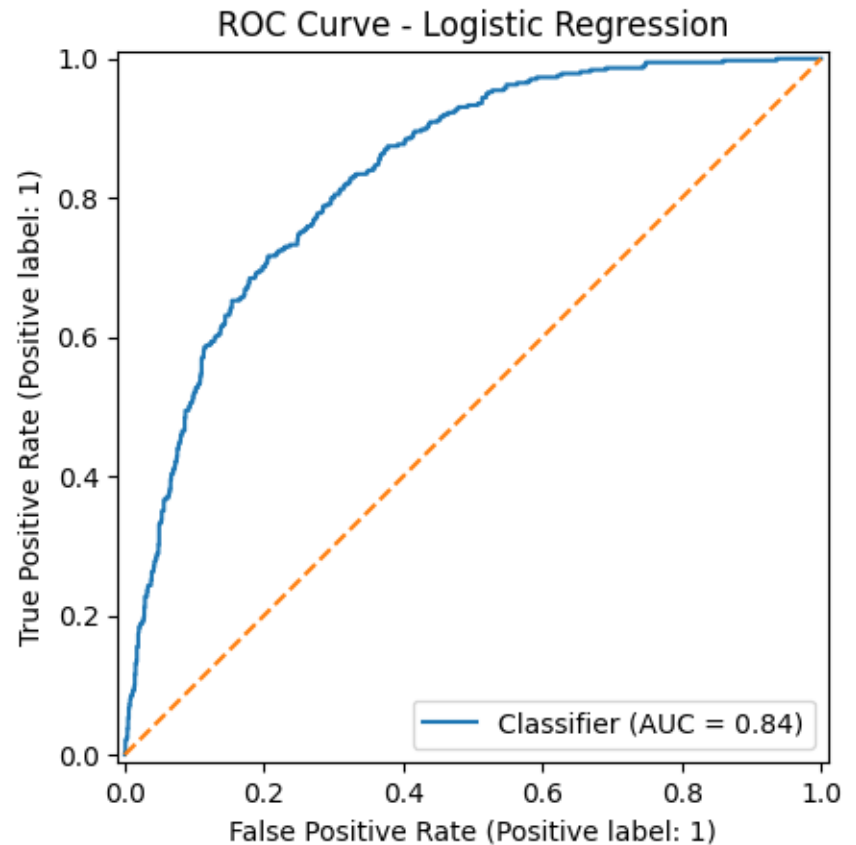
Confusion Matrix Insights

- 916 correctly identified non-churn customers
- 215 correctly identified churn customers
- 159 false negatives (missed churn cases)
- Improving recall increases retention targeting effectiveness



ROC Curve Interpretation

- **ROC-AUC: 0.84**
- Strong discrimination between churn and non-churn
- Model remains above no-skill reference line
- Effective classification across thresholds



Random Forest Comparison

Metric	Logistic	Random Forest
Accuracy	80.4%	78.9%
Precision	0.65	0.63
Recall	0.57	0.52
F1 Score	0.61	0.57
ROC-AUC	0.84	0.82



Logistic Regression outperformed across key metrics

Key Predictors of Churn

- Short tenure increases churn risk
- Month-to-month contracts more likely to churn
- Higher monthly charges associated with attrition
- Fiber optic service linked to churn
- Electronic check payment method increases risk



Business Implications

- Target short-tenure customers with retention offers
- Incentivize longer-term contracts
- Monitor high monthly charge segments
- Develop proactive outreach strategies
- Model supports proactive retention strategy



Conclusion

- Logistic Regression achieved strongest overall performance
- AUC of 0.84 indicates strong model discrimination
- Interpretability supports actionable insights
- Predictive analytics enhances retention strategy

